Programming Languages and Computation

Week 8: Partial Correctness & Loop Invariants

Hoare Logic

* 1. Consider the following *invalid* Hoare triple: $\{x \le y\}\ y \leftarrow y * 2\ \{x \le y\}$. Find an initial state $\sigma \in \mathsf{State}$ and a trace from the configuration $\langle y \leftarrow y * 2, \sigma \rangle$ that contradicts this triple.

Solution

Any state with $2y < x \le y$ suffices, in particular both x and y must be negative.

$$\langle y \leftarrow y * 2, [x \mapsto -3, y \mapsto -2] \rangle \\ \rightarrow [x \mapsto -3, y \mapsto -4]$$

Equivalently, the state is not in $\mathsf{WP}(y \leftarrow y * 2, x \le y) = x \le 2 * y$.

* 2. Find a statement S such that $\{\mathsf{true}\}\ S\ \{z \le x\ \&\&\ z \le y\}$.

Solution

There are multiple possible solutions. One is a program that calculates the minimum:

$$\mathsf{if}\ x \le y\ \{z \leftarrow x\}\ \mathsf{else}\ \{z \leftarrow y\}$$

To check, the weakest pre-condition for the post-condition is $(x \le y \Rightarrow x \le x\ \&\&\ x \le y)\ \&\&\ (!(x \le y) \Rightarrow y \le x\ \&\&\ y \le y)$, which holds in every state.

** 3. We write $d \mid n$ as shorthand for the assertion $\exists k. n = k * d$; that is, n is a multiple of d, or equivalently *d divides n*. For example, $y \mid x$ is satisfied by those states in which the value of x is a multiple of the value of y.

For each of the following statements, compute the weakest pre-condition required to ensure the post-condition $(y \mid x)\ \&\&\ y \ge 0$ holds after execution:

(a) $x \leftarrow x * x$

(b) $\mathsf{if}\ y \le x\ \{x \leftarrow x - y\}\ \mathsf{else}\ \{y \leftarrow y - x\}$

Solution

(a) Substituting $x * x$ for x gives $(y \mid (x * x))\ \&\&\ y \ge 0$.

(b) Any variation of:

$$(y \le x \Rightarrow (y \mid (x-y)) \;\&\&\; y \ge 0) \\ \&\&\;\; (!(y \le x) \Rightarrow ((y-x) \mid x) \;\&\&\; y - x \ge 0)$$

More simply, as $y \mid (x-y)$ exactly when $y \mid x$ (if $x - y = q * y$ then $x = (q+1) * y$, and vice versa), and $!(y \le x)$ means $x < y$ (so $y - x > 0$):

$$(y \le x \Rightarrow (y \mid x) \;\&\&\; y \ge 0) \;\&\&\; (x < y \Rightarrow (y-x) \mid x)$$

** 4. Using your answers to the previous question, for each of the following Hoare triples determine whether they are valid or not. You should justify your answer.

(a) $\{(y \mid x) \;\&\&\; y \ge 0\}\ x \leftarrow x * x\ \{(y \mid x) \;\&\&\; y \ge 0\}$

(b) $\{(y \mid x) \;\&\&\; y \ge 0\}\ x \leftarrow x * x\ \{x \ge y\}$

(c) $\{(y \mid x) \;\&\&\; y \ge 0\}$ if $y \le x$ $\{x \leftarrow x - y\}$ else $\{y \leftarrow y - x\}$ $\{(y \mid x) \;\&\&\; y \ge 0\}$

Solution

(a) This is valid as the pre-condition is included in the weakest pre-condition $(y \mid (x * x)) \;\&\&\; y \ge 0$. In particular, suppose $y \mid x$, i.e. $x = q * y$ for some q, and $y \ge 0$. Then $x * x = x * (q * y) = (x * q) * y$, so $y \mid (x * x)$, and $y \ge 0$ holds by assumption.

(b) This is not valid. The weakest pre-condition of $x \leftarrow x * x$ for the post-condition $x \ge y$ is $x * x \ge y$, and the pre-condition is not included in it. For instance, the state $[x \mapsto 0,\ y \mapsto 1]$ satisfies the pre-condition (as $0 = 0 * 1$), but $0 * 0 \ge 1$ does not hold. Indeed, executing the statement terminates in a state with $x = 0$ and $y = 1$. If we additionally knew that $x \ne 0$, then it would be valid.

(c) This is not valid as the pre-condition is not included in the weakest pre-condition from the previous question:

$$(y \le x \Rightarrow (y \mid x) \;\&\&\; y \ge 0) \;\&\&\; (x < y \Rightarrow (y-x) \mid x)$$

The first conjunct follows from the pre-condition. The second conjunct, however, does not: when $x < y$ we cannot conclude that $y - x$ divides x. For instance, the initial state $[x \mapsto -1,\ y \mapsto 1]$ satisfies the pre-condition (as $-1 = (-1) * 1$) but $x < y$ and $y - x = 2$ does not divide -1. Executing the statement from this state takes the second branch and terminates in $[x \mapsto -1,\ y \mapsto 2]$, which does not satisfy the post-condition as 2 does not divide -1.

** 5. Suppose $S_1, S_2, S_3 \in \mathcal{S}$ are statements satisfying the following Hoare triples:

- $\{x \le y\}\ S_1\ \{y \le x \;\&\&\; x \le 0\}$
- $\{\forall z.\, x * z = 0\}\ S_2\ \{y = z\}$
- $\{x > y \;\&\&\; z = x\}\ S_3\ \{y \le x\}$

Then which of the following triples can be concluded from these? You should briefly justify your answer, e.g. by strengthening pre-conditions, weakening post-conditions, and combining triples.

(a) $\{x \leq y \;\&\&\; x = 0\}\ S_1;\ S_2\ \{y = z\}$

(b) $\{z = x\}$ if $x \leq y$ $\{S_1\}$ else $\{S_3\}$ $\{y \leq x\}$

(c) $\{x \leq y\}\ S_1;$ if $y = 0$ $\{S_2\}$ else $\{y \leftarrow z\}$ $\{y = z\}$

Solution

(a) This triple cannot be concluded as we only know that $x = 0$ prior to the execution of S_1, and S_1 may change x. All we know after S_1 is that $y \leq x \;\&\&\; x \leq 0$. Therefore, we cannot conclude that $\forall z.\, x * z = 0$ (which is equivalent to $x = 0$) holds prior to the execution of S_2.

(b) This triple can be concluded. For the first branch, we have $\{x \leq y \;\&\&\; z = x\}\ S_1\ \{y \leq x\}$, as we can strengthen the pre-condition of the first triple (as $x \leq y \;\&\&\; z = x$ implies $x \leq y$) and weaken its post-condition (as $y \leq x \;\&\&\; x \leq 0$ implies $y \leq x$). For the second branch, $\{!(x \leq y) \;\&\&\; z = x\}\ S_3\ \{y \leq x\}$ is just the third triple, as $!(x \leq y)$ is $x > y$. Both branches have the same post-condition, $y \leq x$, and so that is the post-condition of the conditional.

(c) This triple can be concluded. First, $\{\forall z.\, x * z = 0\}\ S_2\ \{y = z\}$ can be strengthened to $\{x = 0\}\ S_2\ \{y = z\}$, as $x = 0$ implies $\forall z.\, x * z = 0$. For the conditional, we consider the post-condition $y \leq x \;\&\&\; x \leq 0$ of S_1 as a pre-condition. In the first branch, we additionally know $y = 0$, and so $y \leq x \;\&\&\; x \leq 0 \;\&\&\; y = 0$ implies $x = 0$; so S_2 can be executed to give $y = z$. In the second branch, $\mathsf{WP}(y \leftarrow z,\ y = z) = (z = z)$, which is satisfied by every state. Therefore, $\{y \leq x \;\&\&\; x \leq 0\}$ if $y = 0$ $\{S_2\}$ else $\{y \leftarrow z\}$ $\{y = z\}$. Combining this with $\{x \leq y\}\ S_1\ \{y \leq x \;\&\&\; x \leq 0\}$, as the post-condition of the first matches the pre-condition of the second, we conclude the triple as required.

** 6. Let $\mathsf{cmp}_{i,t,f}$ be some statement whose behaviour is determined as follows:

- CMP1. If $\sigma(i) = \sigma(t)$, then $\langle \mathsf{cmp}_{i,t,f}, \sigma \rangle \rightarrow^* \sigma[f \mapsto 1]$;
- CMP2. Otherwise, if $\sigma(i) \neq \sigma(t)$, then $\langle \mathsf{cmp}_{i,t,f}, \sigma \rangle \rightarrow^* \sigma[f \mapsto 0]$.

Let T be the statement $\mathsf{cmp}_{i,t,f};\ i \leftarrow i + 1$.

(a) Determine the state σ such that $\langle T, [i \mapsto 2, t \mapsto 2, f \mapsto 0] \rangle \rightarrow^* \sigma$, justifying your answer in relation to CMP1. and/or CMP2.

(b) Calculate $\mathsf{WP}(T, f = 1 \parallel i \leq t)$ and simplify it as much as you can.

(c) Using your answer to the previous question, determine whether each of the following triples is valid.

$$\{i \leq t\}\ T\ \{f = 1 \parallel i \leq t\} \qquad \{\mathsf{true}\}\ T\ \{f = 1 \parallel i \leq t\}$$

Solution

(a) As $\sigma(i) = 2 = \sigma(t)$ in the initial state, case CMP1. applies and $\mathsf{cmp}_{i,t,f}$ terminates in $[i \mapsto 2, t \mapsto 2, f \mapsto 1]$. Note that i and t are unchanged. The assignment then updates i to $2 + 1 = 3$, and so the final state is $[i \mapsto 3, t \mapsto 2, f \mapsto 1]$.

(b) Working backwards, first through the assignment:

$$\mathsf{WP}(i \leftarrow i + 1, f = 1 \parallel i \leq t) = f = 1 \parallel i + 1 \leq t$$

Then through $\mathsf{cmp}_{i,t,f}$, with one conjunct for each case of its specification:

$$(i = t \Rightarrow 1 = 1 \parallel i + 1 \leq t) \;\&\&\; (i \neq t \Rightarrow 0 = 1 \parallel i + 1 \leq t)$$

The first conjunct is always true and the second simplifies to $i \neq t \Rightarrow i + 1 \leq t$. As we are working with integers, this is equivalent to $i \leq t$. So $\mathsf{WP}(T, f = 1 \parallel i \leq t) = i \leq t$.

(c) The first triple is valid, as the pre-condition $i \leq t$ is exactly the weakest pre-condition. The second is not, as true is not included in $i \leq t$. For instance, $[i \mapsto 5, t \mapsto 3, f \mapsto 0]$ is a counter-example: case CMP2. applies, so $\mathsf{cmp}_{i,t,f}$ terminates with $f = 0$, and then i becomes 6, so we have neither $f = 1$ nor $6 \leq 3$.

*** 7. Strengthening pre-conditions and weakening post-conditions allows us to adapt a Hoare triple to a more general context. However, it is also useful to be able to adapt Hoare triples to contexts with other unrelated variables. This can be done through the *constancy rule*:

$$\frac{\{P\}\ S\ \{Q\}}{\{P \;\&\&\; R\}\ S\ \{Q \;\&\&\; R\}}\ \mathsf{FV}(R) \cap \mathsf{mod}(S) = \emptyset$$

Where $\mathsf{FV}(P)$ for a predicate $P \subseteq \mathsf{State}$ is defined as the set of variables $x \in \mathsf{Var}$ such that $\sigma \in P$ but $\sigma[x \mapsto n] \notin P$ for some state $\sigma \in \mathsf{State}$ and $n \in \mathbb{Z}$; intuitively, those variables whose value effect whether a state is in the set of not and $\mathsf{mod}(S)$ is the set of variables appearing anywhere in the program.

(a) Suppose that S_1 and S_2 are two statement such that $\{P_1\}\ S_1\ \{Q_1\}$ and $\{P_2\}\ S_2\ \{Q_2\}$ where:

- $\mathsf{FV}(P_1), \mathsf{FV}(Q_1) \subseteq \mathsf{mod}(S_1)$;
- $\mathsf{FV}(P_2), \mathsf{FV}(Q_2) \subseteq \mathsf{mod}(S_2)$;
- And $\mathsf{mod}(S_1) \cap \mathsf{mod}(S_2) = \emptyset$.

Justify how $\{P_1 \;\&\&\; P_2\}\ S_1;\ S_2\ \{Q_1 \;\&\&\; Q_2\}$ can be derived from the constancy rule.

(b) Suppose that $P \subseteq \mathsf{State}$ is some set of states. Prove that, if $\sigma \in P$ and $\sigma(x) = \sigma'(x)$ for all $x \in \mathsf{FV}(P)$ and $\#\{x \in \mathsf{Var} \mid \sigma(x) \neq \sigma'(x)\}$ is finite, then $\sigma' \in P$. You may wish to prove it by induction on $\#\{x \in \mathsf{Var} \mid \sigma(x) \neq \sigma'(x)\}$.

(c) Prove that the rule of constancy holds that is:

$$\{P\}\ S\ \{Q\} \Rightarrow \{P \;\&\&\; R\}\ S\ \{Q \;\&\&\; R\}$$

for any statement S, and predicates $P, Q, R \subseteq \mathsf{State}$ such that $\mathsf{FV}(R) \cap \mathsf{mod}(S) = \emptyset$. You may wish to use a result from the previous worksheet.

Solution

(a) By the constancy rule, we have that $\{P_1 \;\&\&\; P_2\}\ S_1\ \{Q_1 \;\&\&\; P_2\}$ as $\mathsf{FV}(P_2) \subseteq \mathsf{mod}(S_2)$ which is disjoint from $\mathsf{mod}(S_1)$. Likewise, we have that $\{Q_1 \;\&\&\; P_2\}\ S_2\ \{Q_1 \;\&\&\; Q_2\}$. Therefore, $\{P_1 \;\&\&\; P_2\}\ S_1; S_2\ \{Q_1 \;\&\&\; Q_2\}$ as required.

(b) Let us write $\mathsf{dis}(\sigma, \sigma')$ for $\#\{x \in \mathsf{Var} \mid \sigma(x) \neq \sigma'(x)\}$.
Let $P \subseteq \mathsf{State}$ be a set of states. We shall prove by induction on n that if:

$$\phi(n): \ \forall \sigma \in P, \sigma' \in \mathsf{State}.\, \mathsf{dis}(\sigma, \sigma') = n \text{ and } \forall x \in \mathsf{FV}(P).\, \sigma(x) = \sigma'(x) \text{ then } \sigma' \in P$$

- In the base case, we have that there are no variables such that $\sigma(x) \neq \sigma'(x)$. Therefore, $\sigma = \sigma'$ and $\sigma' \in P$ by assumption.
- Now suppose $\phi(n)$ holds for some n. Consider two states $\sigma, \sigma' \in \mathsf{State}$ such that $\mathsf{dis}(\sigma, \sigma') = n + 1$ and $\forall x \in \mathsf{FV}(P).\, \sigma(x) = \sigma'(x)$. Let y be some variable such that $\sigma(y) \neq \sigma'(y)$ (which must exists as they disagree on $n + 1$ variables). Note that $y \notin \mathsf{FV}(P)$, else we'd contradict the assumption that $\forall x \in \mathsf{FV}(P).\, \sigma(x) = \sigma'(x)$.
  Now consider the state $\sigma[y \mapsto \sigma'(y)]$. We have that $\mathsf{dis}(\sigma[y \mapsto \sigma'(y)], \sigma') = n$ and $\sigma[y \mapsto \sigma'(y)](x) = \sigma'(x)$ for all $x \in \mathsf{FV}(P)$. Suppose that $\sigma[y \mapsto \sigma'(y)] \notin P$. Then, by definition, $y \in \mathsf{FV}(P)$, which is contradictory. Thus, $\sigma[y \mapsto \sigma'(y)] \in P$. By $\phi(n)$, therefore, $\sigma' \in P$.

(c) From the previous worksheet, we have that, if $\langle S, \sigma\rangle \to^* \sigma'$, then $\sigma(x) = \sigma'(x)$ for all $x \notin \mathsf{mod}(S)$. Now suppose $\sigma \in P \;\&\&\; R$, i.e. it meets the pre-condition. From the rule's premise, we have that $\sigma' \in Q$. To show that $\sigma' \in R$, we must consider the fact that $\mathsf{mod}(S) \cap \mathsf{FV}(R) = \emptyset$. Therefore, $\sigma(x) = \sigma'(x)$ for all $x \in \mathsf{FV}(R)$. As σ and σ' can only differ on variables in S, by the previous exercise, $\sigma' \in R$. Therefore, $\sigma' \in P \;\&\&\; R$ as required.

Loop Invariants

Recall that an assertion I is a *loop invariant* for $\mathsf{while}\ e\ \{S\}$ if $\{I \;\&\&\; e\}\ S\ \{I\}$, i.e. $I \;\&\&\; e \subseteq \mathsf{WP}(S, I)$. To show that $\{P\}\ \mathsf{while}\ e\ \{S\}\ \{Q\}$ holds using an invariant I, we additionally need $P \subseteq I$, i.e. the invariant implies the pre-condition, and $I \;\&\&\; !\,e \subseteq Q$, i.e. the invariant implies the post-condition on exit.

** 8. For each of the following statements, determine whether $q \leq x$ is a loop invariant. If so, justify why. Otherwise, provide a counter-example.

(a) $\mathsf{while}\ y * q \leq x\ \{q \leftarrow q + 1\}$

(b) $\mathsf{while}\ !(q \leq x)\ \{x \leftarrow x - 1\}$

(c) $\mathsf{while}\ !(x \leq q * q)\ \{q \leftarrow q + 1\}$

Solution

(a) $q \leq x$ is not a loop invariant. We have $\mathsf{WP}(q \leftarrow q + 1, q \leq x) = q + 1 \leq x$. Consider the state $[\,]$ where all variables are mapped to 0. It satisfies the branch condition $y * q \leq x$ and the candidate invariant, but not $q + 1 \leq x$. Indeed, we have $\langle q \leftarrow q + 1, [\,]\rangle \to [q \mapsto 1]$ and thus the candidate loop invariant is not preserved.

(b) $q \leq x$ is a loop invariant. For it to be a loop invariant, it must be the case that $q \leq x$ && $!(q \leq x) \subseteq \mathsf{WP}(x \leftarrow x-1, q \leq x)$. The left-hand side is unsatisfiable, and so the inclusion holds trivially. In other words, when $q \leq x$, the loop is never entered and so the condition is trivially preserved.

(c) $q \leq x$ is a loop invariant. We have $\mathsf{WP}(q \leftarrow q+1, q \leq x) = q+1 \leq x$ and so we must show that $q \leq x$ && $!(x \leq q*q)$ implies $q+1 \leq x$. Suppose $x > q*q$. For any integer q we have $q*q \geq q$ (as $q*q-q = q*(q-1)$ is a product of consecutive integers, and so is non-negative). Therefore, $x > q*q \geq q$ and so $x \geq q+1$ as we are working with integers, as required.

** 9. Find a loop invariant that can be used to demonstrate the following Hoare triple. You should justify your solution by checking that the invariant is implied by the pre-condition (after the initialisation), preserved by the loop body, and implies the post-condition, calculating weakest pre-conditions as necessary.

$\{0 \leq x$ && $0 < y\}$
$q \leftarrow 0;$
$r \leftarrow x;$
while $y \leq r$ {
  $r \leftarrow r-y;$
  $q \leftarrow q+1$
}
$\{x = q*y+r$ && $r < y\}$

Solution

An invariant for this loop is $I := x = q*y+r$.

- **Implied by the pre-condition.** Working backwards through the initialisation, $\mathsf{WP}(r \leftarrow x, I) = x = q*y+x$ and $\mathsf{WP}(q \leftarrow 0, x = q*y+x) = x = 0*y+x$. This holds in every state, and so is implied by $0 \leq x$ && $0 < y$.
- **Preserved by the loop body.** We have $\mathsf{WP}(q \leftarrow q+1, I) = x = (q+1)*y+r$, and then $\mathsf{WP}(r \leftarrow r-y, x = (q+1)*y+r) = x = (q+1)*y+(r-y)$. This is equivalent to $x = q*y+r$, i.e. I itself, and so I && $y \leq r \subseteq \mathsf{WP}(\text{body}, I)$.
- **Implies the post-condition.** We have that I && $!(y \leq r)$ is $x = q*y+r$ && $r < y$, which is exactly the post-condition.

*** 10. Find a loop invariant that can be used to demonstrate the following Hoare triple. You should justify your solution by checking that the invariant is implied by the pre-condition (after the initialisation), preserved by the loop body, and implies the post-condition, calculating weakest pre-conditions as necessary.

$$
\begin{array}{l}
\{n \geq 0\} \\
i \leftarrow 0; \\
s \leftarrow 0; \\
\text{while } !(i = n) \ \{ \\
\quad s \leftarrow s + 2 * i + 1; \\
\quad i \leftarrow i + 1 \\
\} \\
\{s = n * n\}
\end{array}
$$

Solution

An invariant is $I := s = i * i$ (this can be hypothesized by trialling a few iterations, or observing that $(x+1)^2 = x^2 + 2x + 1$).

- **Implied by the pre-condition.** We have $\mathsf{WP}(s \leftarrow 0, I) = 0 = i * i$, and then $\mathsf{WP}(i \leftarrow 0, 0 = i * i) = 0 = 0 * 0$. This holds in every state, and so is implied by $n \geq 0$.
- **Preserved by the loop body.** We have $\mathsf{WP}(i \leftarrow i + 1, I) = s = (i+1) * (i+1)$, and then $\mathsf{WP}(s \leftarrow s + 2 * i + 1, s = (i+1) * (i+1)) = s + 2 * i + 1 = (i+1) * (i+1)$. As $(i+1) * (i+1) = i * i + 2 * i + 1$, this is equivalent to $s = i * i$, i.e. I itself. So I && $!(i = n) \subseteq \mathsf{WP}(\text{body}, I)$.
- **Implies the post-condition.** If $s = i * i$ and $i = n$, then $s = n * n$ as required.

*** 11. Let $\mathsf{cmp}_{i,t,f}$ be the statement from the earlier exercise, satisfying CMP1. and CMP2. Consider the following program, which we shall refer to as S, that searches for the value of t amongst the numbers from 1 up to n.

$$
\begin{array}{l}
i \leftarrow 1; \\
f \leftarrow 0; \\
\text{while } i \leq n \ \&\& \ f = 0 \ \{ \\
\quad \mathsf{cmp}_{i,t,f}; \\
\quad i \leftarrow i + 1 \\
\}
\end{array}
$$

(a) Determine the final state σ such that $\langle S, [n \mapsto 5, t \mapsto 3] \rangle \rightarrow^* \sigma$ and justify your answer in relation to CMP1. and/or CMP2.

(b) Find a loop invariant that justifies the following partial correctness property and prove that it is valid by showing that it is implied by the pre-condition (after the initialisation), preserved by the loop body, and implies the post-condition.

$$\{1 \leq t \ \&\& \ t \leq n\} \ S \ \{f = 1\}$$

You should make reference to CMP1. and CMP2. in your proof.

Solution

(a) After the initialisation, the state is $[n \mapsto 5, t \mapsto 3, i \mapsto 1, f \mapsto 0]$.

- As $i = 1 \leq 5$ and $f = 0$, the loop body executes. As $\sigma(i) = 1 \neq 3 = \sigma(t)$, case CMP2. applies and f is set to 0; then i becomes 2.
- As $i = 2 \leq 5$ and $f = 0$, the loop body executes. As $2 \neq 3$, case CMP2. applies again and f remains 0; then i becomes 3.
- As $i = 3 \leq 5$ and $f = 0$, the loop body executes. As $\sigma(i) = 3 = \sigma(t)$, case CMP1. applies and f is set to 1; then i becomes 4.
- The branch condition $i \leq n$ && $f = 0$ no longer holds as $f = 1$, and so the loop terminates.

Therefore, the final state is $[n \mapsto 5, t \mapsto 3, i \mapsto 4, f \mapsto 1]$.

(b) An invariant is $I := t \leq n$ && $(f = 1 \,\|\, (f = 0$ && $i \leq t))$: either the target has already been found, or it has not been and the counter has not yet gone past it. The first conjunct is not changed by the loop, but we will need it to show that the post-condition holds.

- **Implied by the pre-condition.** We calculate $\mathsf{WP}(f \leftarrow 0, I) = t \leq n$ && $(0 = 1 \,\|\, (0 = 0$ && $i \leq t))$, which is equivalent to $t \leq n$ && $i \leq t$. Then $\mathsf{WP}(i \leftarrow 1, t \leq n$ && $i \leq t) = t \leq n$ && $1 \leq t$, which is exactly the pre-condition.
- **Preserved by the loop body.** We work backwards.

$$\begin{aligned} \mathsf{WP}(i \leftarrow i+1, I) &= t \leq n \;\&\&\; (f = 1 \,\|\, (f = 0 \;\&\&\; i+1 \leq t)) \\ \mathsf{WP}(\mathsf{cmp}_{i,t,f}, \cdots) &= (i = t \Rightarrow t \leq n \;\&\&\; (1 = 1 \,\|\, \cdots)) \\ &\quad \&\& \; (i \neq t \Rightarrow t \leq n \;\&\&\; (0 = 1 \,\|\, (0 = 0 \;\&\&\; i+1 \leq t))) \end{aligned}$$

  which simplifies to $(i = t \Rightarrow t \leq n)$ && $(i \neq t \Rightarrow t \leq n$ && $i + 1 \leq t)$. Now suppose I && $i \leq n$ && $f = 0$. As $f = 0$ we must have $i \leq t$, and we have $t \leq n$. If $i = t$, then the first conjunct holds as $t \leq n$. Otherwise, $i \neq t$ and so $i < t$, which gives $i + 1 \leq t$ as we are working with integers; together with $t \leq n$ the second conjunct holds. Intuitively, the first case corresponds to CMP1. (the target is found and so f becomes 1) and the second to CMP2. (the target is not found, but we haven't yet gone past it).
- **Implies the post-condition.** Suppose I holds and the branch condition fails, i.e. $!(i \leq n$ && $f = 0)$. If $f = 1$, then the post-condition holds. Otherwise, by I, we have $f = 0$ and $i \leq t$. As $f = 0$, the branch condition failing means $i > n$. But $i \leq t \leq n$, which is a contradiction, so this case cannot occur.

*** 12. Find a pair of loop invariants (one for each loop) that together can be used to demonstrate the following Hoare triple. You should justify your solution by checking, for each loop, the three constraints on a loop invariant, calculating weakest pre-conditions as necessary.

$\{0 \leq n\}$
$i \leftarrow 0;$
$j \leftarrow 0;$
while $i \leq n - 1$ {
  $j \leftarrow 0;$
  while $j \leq i$ {
    $j \leftarrow j + 1$
  };
  $i \leftarrow i + 1$
}
$\{j = n\}$

Solution

Let $J := 0 \leq i$ && $i \leq n$ && $j = i$ be the invariant for the outer loop, and $K := 0 \leq j$ && $j \leq i+1$ && $i+1 \leq n$ be the invariant for the inner loop. The body of the outer loop is $j \leftarrow 0;\ L;\ i \leftarrow i+1$, where L is the inner loop.

Inner loop.

- **Preserved by the loop body.** $\mathsf{WP}(j \leftarrow j+1, K) = 0 \leq j+1$ && $j+1 \leq i+1$ && $i+1 \leq n$. Suppose K && $j \leq i$. Then $0 \leq j+1$ as $0 \leq j$, $j+1 \leq i+1$ as $j \leq i$, and $i+1 \leq n$ is part of K.
- **Implies the post-condition.** What follows the inner loop is $i \leftarrow i+1$ and we need J afterwards, and so we need K && $!(j \leq i) \subseteq \mathsf{WP}(i \leftarrow i+1, J)$, where $\mathsf{WP}(i \leftarrow i+1, J) = 0 \leq i+1$ && $i+1 \leq n$ && $j = i+1$. From K, we have $j \leq i+1$ and from $!(j \leq i)$ we have $j > i$, and so $j = i+1$. Then $0 \leq i+1$ as $0 \leq j$, and $i+1 \leq n$ is part of K.
- Therefore $K \subseteq \mathsf{WP}(L, \mathsf{WP}(i \leftarrow i+1, J))$.

Outer loop.

- **Implied by the pre-condition.** $\mathsf{WP}(i \leftarrow 0;\ j \leftarrow 0, J) = 0 \leq 0$ && $0 \leq n$ && $0 = 0$, which is equivalent to $0 \leq n$.
- **Preserved by the loop body.** We have that $\mathsf{WP}(j \leftarrow 0, K) = 0 \leq 0$ && $0 \leq i+1$ && $i+1 \leq n$. Suppose J && $i \leq n-1$. Then $0 \leq i$ and $i+1 \leq n$ and so this holds. By the inner loop above and monotonicity, $\mathsf{WP}(j \leftarrow 0, K) \subseteq \mathsf{WP}(j \leftarrow 0;\ L;\ i \leftarrow i+1, J)$, which is the weakest pre-condition of the outer loop body for J.
- **Implies the post-condition.** J && $!(i \leq n-1)$ gives $i \leq n$ and $i > n-1$, so $i = n$, and as $j = i$ we have $j = n$ as required.

** 13. Find a loop invariant for the following program that demonstrates that this program has at least one non-terminating trace. You should justify your answer in relation to the definition of a Hoare triple.

$$\text{while } x > 0 \ \{x \leftarrow x + y\}$$

Solution

A possible invariant is simply $I := y \geq 0$ && $x > 0$. To verify this, we need to show that I && $x > 0 \subseteq \mathsf{WP}(x \leftarrow x+y, I)$. We have $\mathsf{WP}(x \leftarrow x+y, I) = y \geq 0$ && $x+y > 0$, which is indeed implied by $y \geq 0$ && $x > 0$.

Moreover, I && $!(x > 0)$ is unsatisfiable, and so is included in false. This allows us to conclude that $\{y \geq 0$ && $x > 0\}\ S\ \{\text{false}\}$ where S is the above program. By the definition of a Hoare triple, we have that if σ satisfies $y \geq 0$ && $x > 0$ and $\langle S, \sigma\rangle \rightarrow^* \sigma'$ then σ' satisfies false. In other words, in any state σ such that $\sigma(y) \geq 0$ and $\sigma(x) > 0$, there is no terminal state, and so S has non-terminating traces. (Note that the weaker pre-condition $y \geq 0$ would not suffice, as when $x \leq 0$ the loop exits immediately.)